

IZYUM, Ukraine

WMO: 344150

Lat: 49.1872N

Lon: 37.2894E

Elev: 256

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	0.4	-11.5	3.0	-6.2	-5.2	4.2	0.9	17.2	26.3	15.0	26.7	2.6	340	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	92.7	69.0	89.1	67.8	86.1	67.1	71.8	85.4	70.3	83.9	69.0	82.0	5.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	102.1	76.8	66.0	96.8	75.5	64.6	92.2	73.8	35.7	85.6	34.4	83.9	33.3	82.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.8	13.7	11.5	DB	-14.8	97.0	6.6	4.2	-19.5	100.0	-23.4	102.4	-27.2	104.8	-32.0	107.8	(4)
			WB	-15.0	75.2	6.6	2.7	-19.8	77.1	-23.6	78.7	-27.3	80.2	-32.1	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	23.9	25.9	35.5	49.7	60.8	68.0	71.7	69.4	58.8	46.7	36.6	27.9	(6)
		DBStd	19.01	11.57	11.61	8.20	7.22	7.65	6.34	5.46	6.28	7.44	8.32	9.00	10.75	(7)
		HDD50	3306	810	675	451	90	9	0	0	14	163	409	685		(8)
		HDD65	6773	1275	1095	914	459	174	39	8	28	211	568	852	1150	(9)
		CDD50	2588	0	0	2	82	343	541	673	601	277	61	8	0	(10)
		CDD65	579	0	0	0	1	43	130	216	164	24	1	0	0	(11)
		CDH74	6865	0	0	0	30	617	1428	2344	2065	366	15	0	0	(12)
		CDH80	2598	0	0	0	3	176	487	951	888	92	1	0	0	(13)
(14)	Wind	WSAvg	4.4	4.6	5.1	5.5	4.9	4.3	4.1	3.6	3.4	3.6	3.9	4.7	5.0	(14)
(15)	Precipitation	PrecAvg	22.4	1.8	1.5	1.7	1.6	2.0	2.9	1.9	1.5	2.0	1.7	1.6	2.0	(15)
(16)		PrecMax	37.6	3.3	4.3	3.4	4.8	5.1	13.7	6.1	5.9	4.2	3.1	4.8	4.7	(16)
(17)		PrecMin	15.7	0.6	0.4	0.4	0.2	0.2	0.5	0.2	0.2	0.0	0.4	0.0	0.7	(17)
(18)		PrecStd	4.8	0.8	0.9	0.9	1.1	1.1	2.5	1.3	1.1	1.3	0.8	1.1	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	46.0	50.6	62.6	78.0	89.9	92.3	97.2	97.7	87.6	75.5	61.8	50.9		(19)
(20)		0.4% MCWB	43.7	46.4	50.7	57.3	65.3	68.8	69.5	69.2	65.2	60.9	55.1	48.2		(20)
(21)		2% DB	42.1	44.4	56.3	72.3	84.7	88.8	92.9	93.0	82.5	69.2	56.8	46.7		(21)
(22)		2% MCWB	40.5	41.4	46.0	55.5	62.7	67.8	69.7	68.3	62.8	58.0	52.4	44.6		(22)
(23)		5% DB	39.4	41.0	51.9	68.5	81.0	85.7	88.8	88.8	78.3	63.7	52.5	43.3		(23)
(24)		5% MCWB	38.0	38.6	44.3	53.5	62.5	67.2	68.7	67.3	61.1	56.4	49.2	41.7		(24)
(25)		10% DB	36.7	38.5	47.5	64.1	76.7	82.2	85.5	85.1	73.8	59.6	48.7	40.3		(25)
(26)		10% MCWB	35.4	36.6	42.3	51.3	60.9	65.8	68.2	66.6	60.1	54.2	46.2	39.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	44.6	47.2	51.1	59.6	68.6	72.6	74.0	73.0	68.4	62.8	56.0	48.5		(27)
(28)		0.4% MCDB	46.0	50.1	60.4	73.9	83.8	85.8	87.4	87.7	82.8	69.4	59.5	50.3		(28)
(29)		2% WB	40.7	41.9	47.6	57.2	65.9	70.8	72.0	71.1	65.5	59.8	52.9	44.8		(29)
(30)		2% MCDB	41.7	43.8	53.6	68.7	78.8	83.5	85.3	86.0	77.3	66.2	56.3	46.6		(30)
(31)		5% WB	38.0	39.0	45.0	55.2	64.2	69.1	70.7	69.3	63.2	57.1	49.5	41.9		(31)
(32)		5% MCDB	39.1	40.6	50.2	64.6	75.7	81.2	83.9	83.9	73.3	62.6	52.0	43.1		(32)
(33)		10% WB	35.5	36.8	42.5	53.3	62.5	67.4	69.4	67.8	61.3	54.5	46.3	39.0		(33)
(34)		10% MCDB	36.3	38.3	47.3	61.7	73.5	78.7	82.2	81.1	70.8	59.3	48.2	40.2		(34)
(35)	Mean Daily Temperature Range	MDBR	9.7	11.4	13.9	20.2	22.5	21.5	22.2	24.9	22.8	17.1	10.1	8.9		(35)
(36)		5% DB MCDBR	7.6	12.2	22.5	28.5	29.0	26.7	27.9	31.8	30.6	25.6	14.6	9.2		(36)
(37)		5% DB MCWBR	7.0	9.7	13.5	13.5	10.9	9.5	9.0	10.5	13.4	14.6	10.7	8.2		(37)
(38)		5% WB MCDBR	7.3	11.2	18.9	23.0	23.6	22.6	23.4	26.3	25.1	21.2	12.8	8.9		(38)
(39)		5% WB MCWBR	6.9	9.4	12.6	11.8	10.3	9.4	9.3	10.4	12.7	13.1	10.1	8.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.358	0.390	0.462	0.419	0.413	0.443	0.441	0.411	0.375	0.355	0.332		(40)
(41)		taud	2.153	2.081	2.143	2.043	2.258	2.301	2.224	2.216	2.265	2.326	2.284	2.221		(41)
(42)		Ebn at Noon	218	240	255	248	267	269	259	252	247	236	210	200		(42)
(43)		Edh at Noon	26	36	40	49	41	40	43	41	35	28	23	22		(43)
(44)	All-Sky Solar Radiation	RadAvg	300	595	954	1367	1826	1952	1930	1723	1187	684	303	191		(44)
(45)		RadStd	60	115	137	137	143	163	131	126	136	110	62	33		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	+1.44	N/A	N/A	+2.02	N/A	N/A	-275	-401	+234	+139

Nomenclature: See separate page